

Shiv Nadar Institution of Eminence Deemed to be University, Delhi NCR

Department of Computer Science and Engineering

Course: Software Engineering (CSD326)

Lab Assignment 2: Data Flow Diagram (DFD) & Data Dictionary

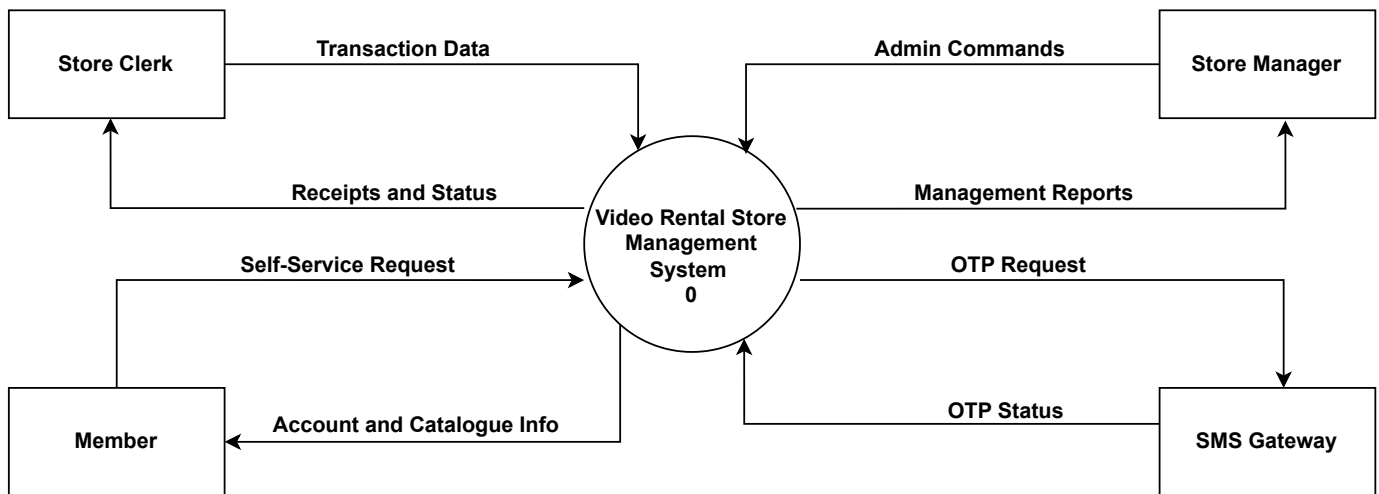
Project: Video Rental Store Management System (VRSMS)

Submitted by: Group 23

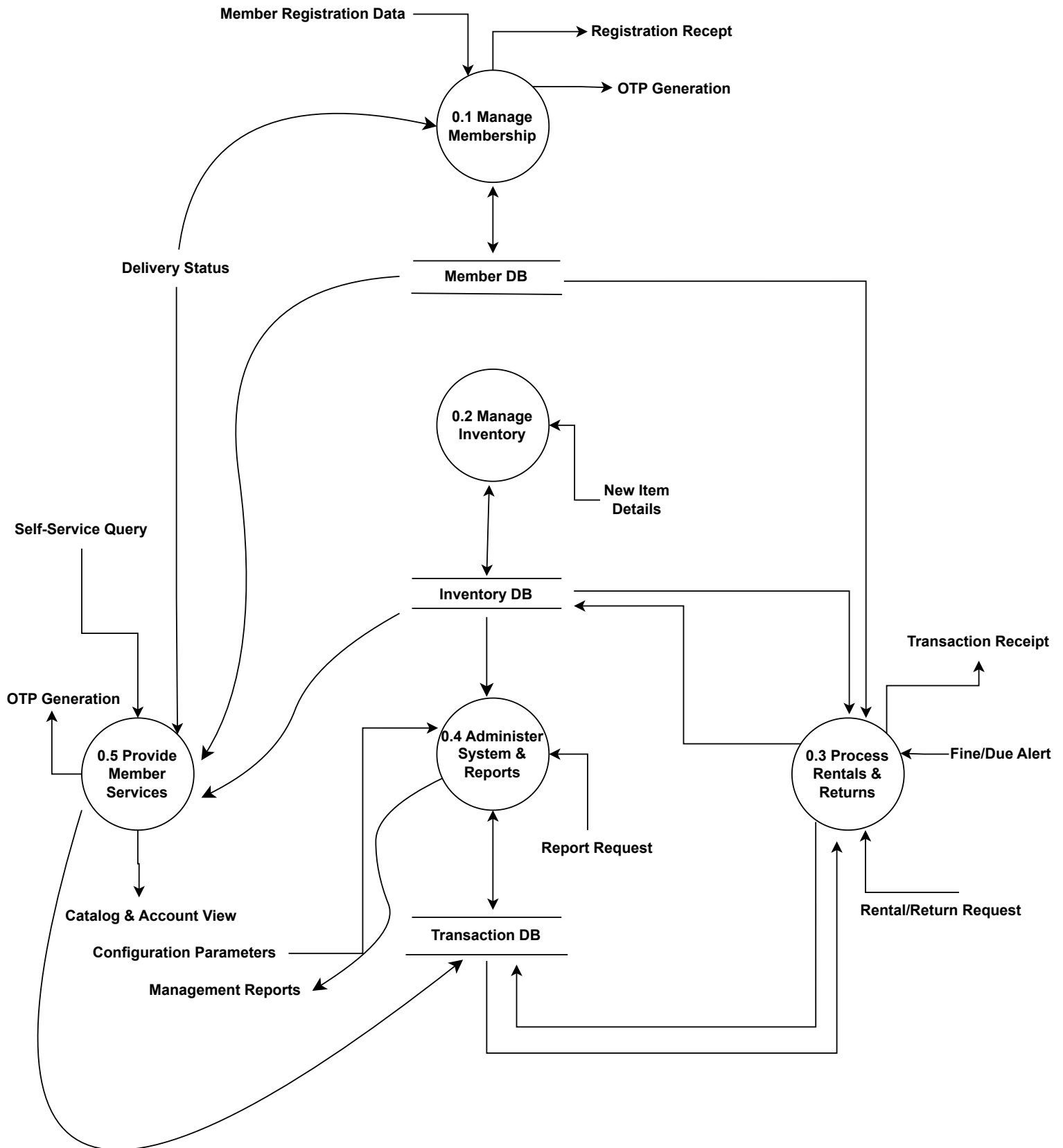
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Due Date: February 17, 2026

Context Diagram

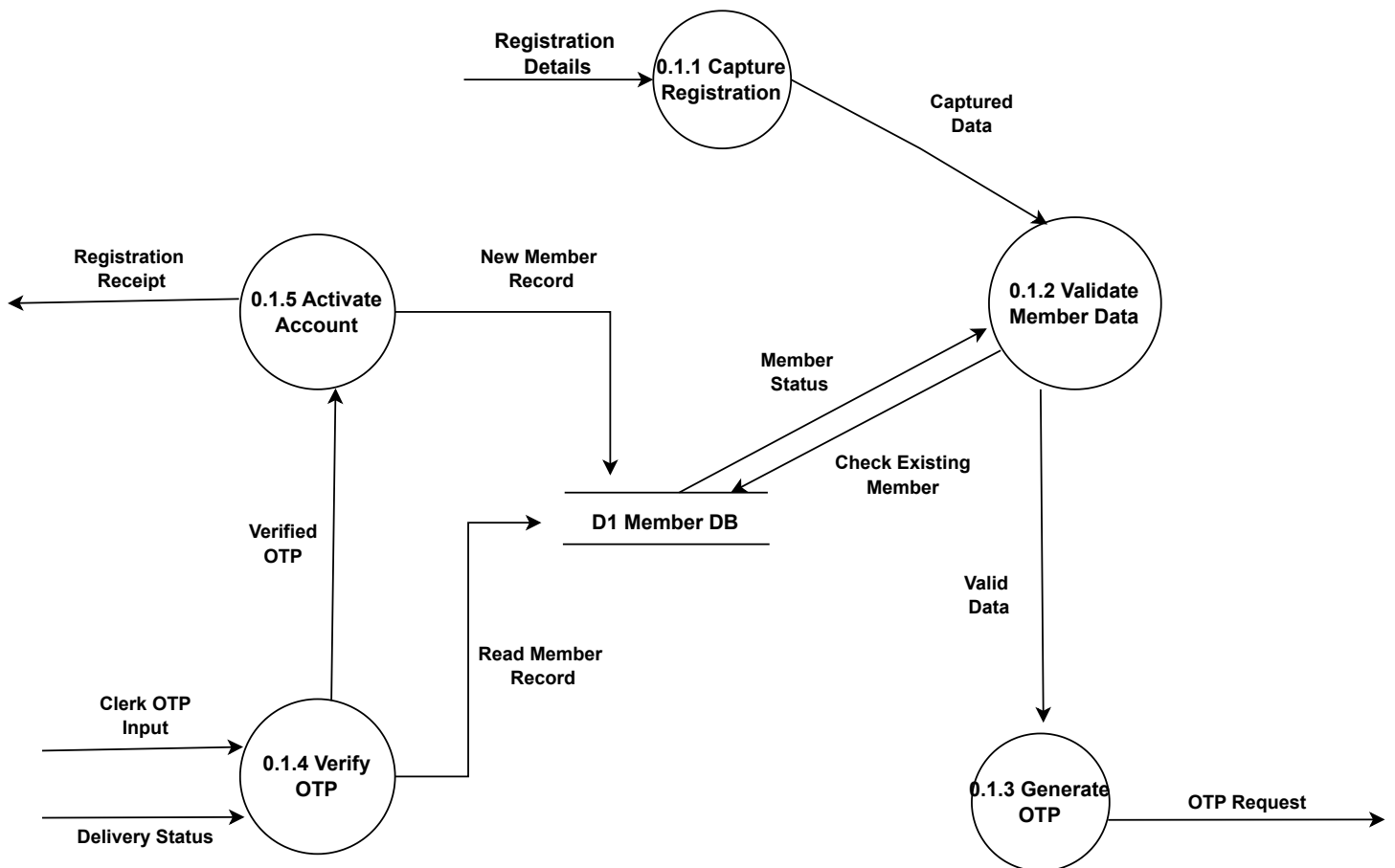


LEVEL 1 Data Flow Diagram



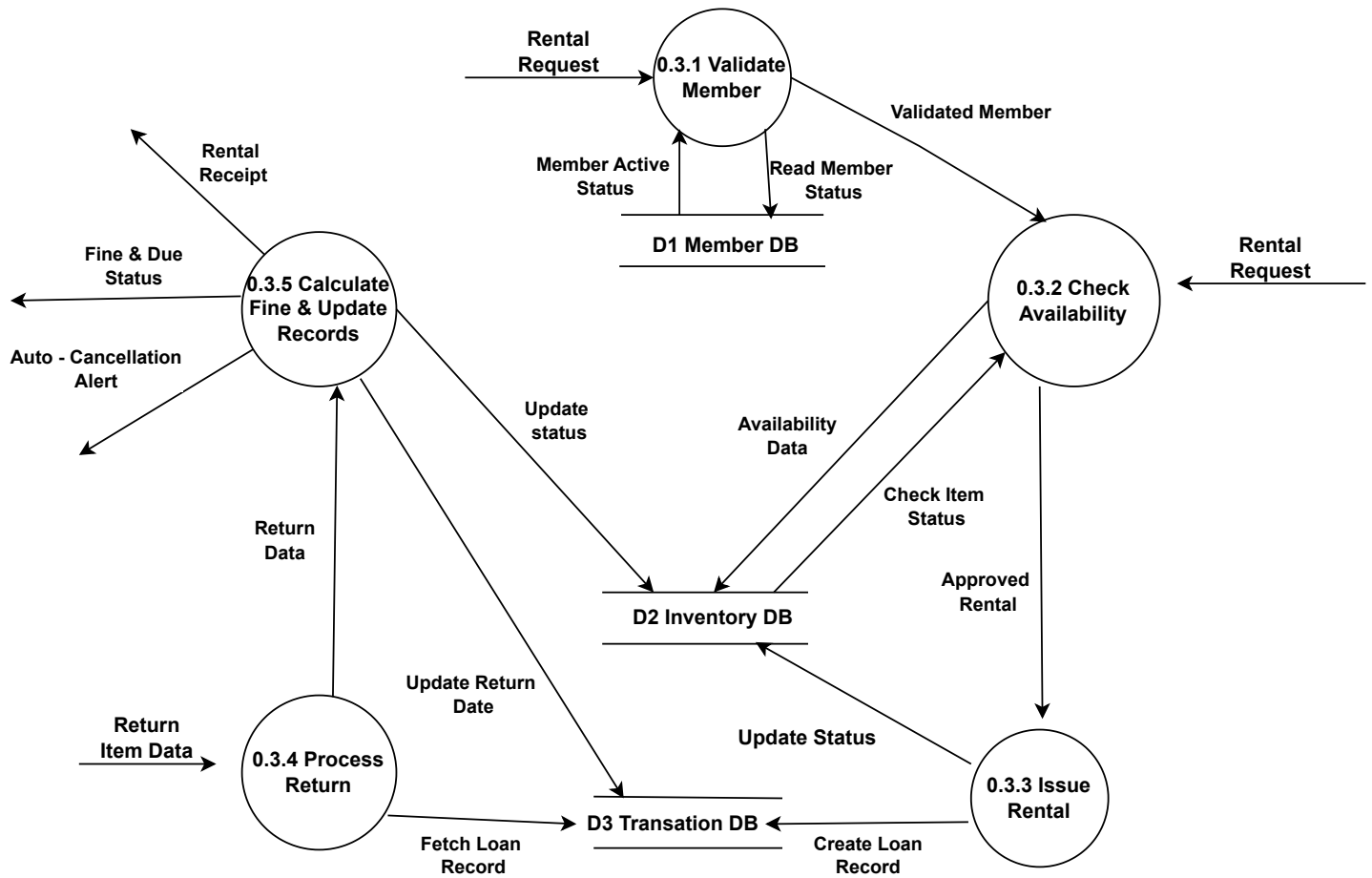
LEVEL 2 Data Flow Diagram

Manage Membership



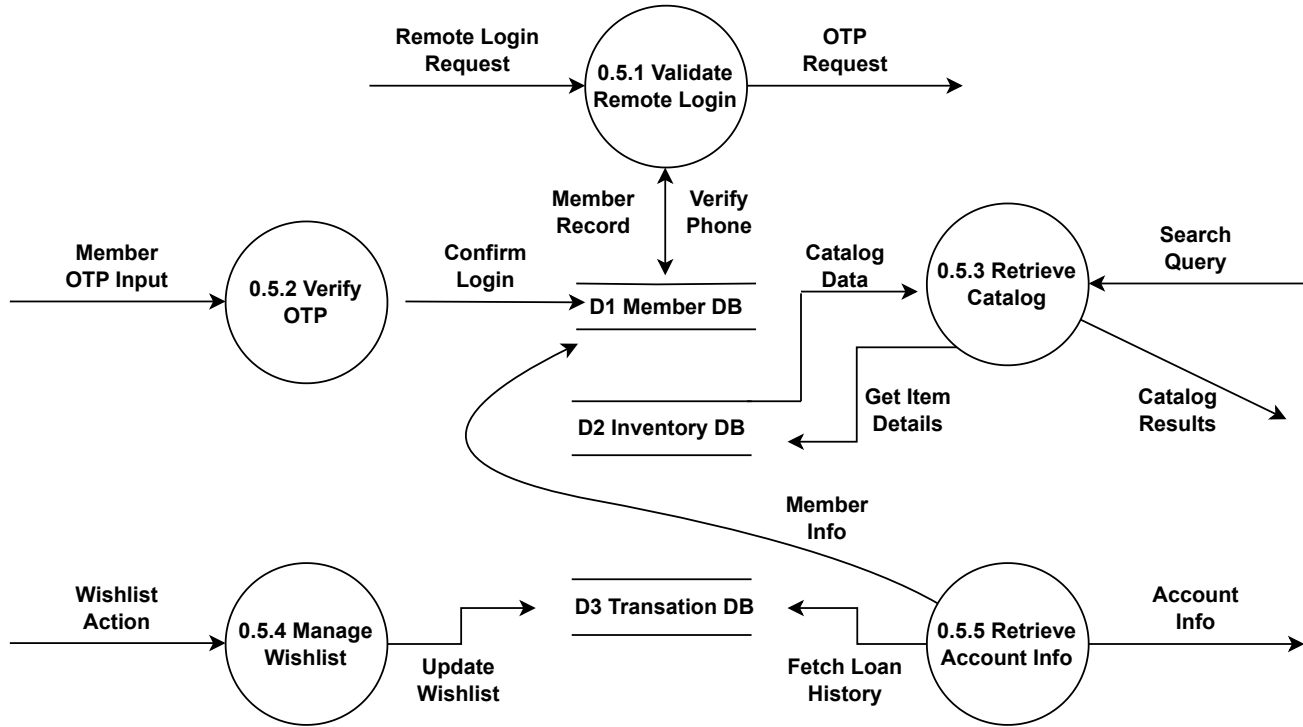
LEVEL 2 Data Flow Diagram

Process Rentals



LEVEL 2 Data Flow Diagram

Provide Member Services



Data Dictionary

Transaction Data: Member Registration Data + Rental/Return Request + Payment Confirmation + Clerk OTP Input

Member Registration Data: Name + Address + Phone Number + Photo Data + Deposit Amount

Rental/Return Request: Member ID + Item ID + Transaction Type [Issue — Return]

Payment Confirmation: Transaction ID + Amount + Payment Mode [Cash — UPI] + Status

Clerk OTP Input: OTP Code

Receipts and Status: Registration Receipt + Transaction Receipt + Fine/Due Alert + Success/Error Message

Registration Receipt: Member ID + Name + Date + Deposit Amount

Transaction Receipt: Transaction ID + Member ID + Item Details + Date + Due Date + Amount Paid

Fine/Due Alert: Member ID + Amount Due + Reason [Overdue — Damage]

Admin Commands: New Item Details + Config Parameters + Report Request

New Item Details: Title + Category [Video — Music] + Format [VHS — MP4 — CD] + Purchase Price + Purchase Date

Config Parameters: Max Loan Duration + Overdue Fine Amount + Cancellation Threshold

Report Request: Report Type [Profit/Loss — Dead Stock] + Date Range

Management Reports: Profit/Loss Data + Dead Stock Report + Auto-Cancellation Alert

Profit/Loss Data: Total Income (Rentals + Fines) - Total Expenses (Purchases)

Dead Stock Report: {Item ID + Title + Last Issue Date}

Self-Service Request: Remote Login Request + Search Query + Wishlist Action + Member OTP Input

Remote Login Request: Phone Number

Search Query: Keywords + Filters [Category — Genre — Format]

Wishlist Action: Item ID + Action [Add — Remove]

Member OTP Input: OTP Code

Account and Catalogue Info: Catalog Results + Account Info + OTP Request

Catalog Results: {Item ID + Title + Status [Available — On Loan] + Due Date}

Account Info: Loan History + Wishlist View + Current Dues

Loan History: {Item ID + Issue Date + Return Date + Fine Amount}

Current Dues: Total Rent + Total Fines

OTP Request: Phone Number + Generated OTP Code

OTP Status: Delivery Status [Success — Failure]

Member Record: Member ID + Name + Address + Phone + Photo + Registration Date + Status [Active — Cancelled]

Item Record: Item ID + Title + Category + Format + Purchase Price + Daily Rate + Status

Loan Record: Loan ID + Member ID + Item ID + Issue Date + Expected Return Date + Actual Return Date + Fine Amount